Notice of Partnering Opportunity
Lunar Dust Reduction through Electrostatic Adhesion Mitigation (L-DREAM)
80GRC025F7020-NPO

General Information

NASA's Space Technology Mission Directorate is seeking proposals for the Early Career Initiative (ECI), a funding opportunity to engage NASA early career researchers with world class partners to develop the innovative leaders and technologies of the future. The ECI proposal titled, "Lunar Dust Reduction through Electrostatic Adhesion Mitigation (L-DREAM)" has been nominated by NASA Glenn Research Center for final selection by NASA Headquarters. L-DREAM seeks to develop a lunar dust mitigation coating which minimizes electrostatic adhesion forces and can be applied onto solar cells and thermal control surfaces to reduce performance losses due to lunar dust. L-DREAM ECI proposers are seeking a partner organization external to NASA capable of providing services relating to developing this coating for compatibility with large-scale manufacturing methods.

Purpose of Notice

NASA Glenn Research Center (GRC) is issuing this Notice of Partnering Opportunity (NPO) for the purpose of openly and competitively selecting prospective team members to assist it in the preparation of proposals to be submitted by GRC to the Early Career Initiative. This process is being conducted in accordance with NASA FAR Supplement 1872.3. By this Notice, GRC is seeking information from partners regarding their availability, cost, capabilities, and past performance that relates to the science partnerships on potential future missions, technology and instrument developments, and laboratory experiments. This Notice also authorizes NASA GRC to award grants, cooperative agreements and/or contracts to its selected partner(s) without further competition or justification, for execution of any GRC proposal that is selected in these partnerships.

Overview

This Notice is not to be construed as a commitment by the Government to enter into a partnership. This Notice is subject to revision or cancellation at any time. Based on the responses to this Notice, GRC anticipates selecting organizations with which GRC would then immediately work directly and individually to develop, complete, and submit proposals in response to the above-mentioned objectives.

During the development of specific proposals, specific roles and cost estimates for the selected partners will be developed. For any GRC proposal that is selected, GRC intends to negotiate, without further competition or justification, a grant, cooperative agreement, or contract with each proposal partner, consistent with their agreed upon roles and this Notice. During these procurement negotiations, the specific statement of work, terms and conditions, and procurement value will be determined. There will be no exchange of funds between GRC and the teaming partners during the preparation of proposals. NASA GRC will not be responsible for any costs incurred by respondents in preparing a response to proposals or by the selected partners in assisting GRC in the preparation of proposals.

SECTION 1—INTRODUCTION

General Information

On the Moon, sustained operation of power systems, habitats, rovers, and astronauts will be challenging without mitigating lunar dust. Lunar dust consists of tiny, abrasive, electrically charged particles that reduce visibility, alter optical properties, and inhibit operations of mission critical components. The Moon's intense charging environment can charge dust, making it difficult to overcome electrostatic adhesion forces to remove dust from surfaces. When dust settles onto solar cells the opaque dust particles reduce the transmittance of light into the solar cell, degrading power output. Thermal control surfaces rely on reflecting incident radiation to keep underlying surfaces cool. When dust settles onto thermal control surfaces, the solar absorptance increases, which causes them to heat up more from incident radiation. Existing passive dust mitigation solutions are few and have relatively low Technology Readiness Level (TRL). The L-DREAM proposal plans to address this technology gap by providing a passive dust mitigation coating which can minimize electrostatic adhesion forces. Most of the coating development and testing work will be completed internally by NASA. However, successful infusion of this technology into Artemis program elements will require development of the capability to mass-manufacture the coating and deposit it at a large scale, an objective suitable for an external partner.

Deliverables expected of the external partner during the L-DREAM ECI project performance period include:

- Develop magnetron sputter deposition parameters for large scale manufacturing of L-DREAM coating.
- Iteratively deposit L-DREAM coating using magnetron sputter deposition onto substrates including (a) solar cell coverglass or flexible coverglass and (b) thermal control surface materials. Substrates and sputter targets will be provided by NASA.
- Collaborate with NASA to refine L-DREAM coating and manufacturing methods to achieve target coating properties.

Participation

Participation in this partnering synopsis is open to all categories of U.S. and non-U.S. organizations, including educational institutions, industry, not-for-profit institutions, the Jet Propulsion Laboratory, as well as NASA Centers and other U.S. Government Agencies. Historically Black Colleges and Universities (HBCUs), Other Minority Universities (OMUs), small, disadvantaged businesses (SDBs)(8a), veteran-owned small businesses, service, disabled veteran-owned small businesses, HUBZone small businesses, and women-owned small businesses (WOSBs) are encouraged to apply. Participation by non-U.S. organizations is welcome but subject to NASAs policy of no exchange of funds, in which each government supports its own national participants and associated costs.

SECTION 2—SCOPE

Overview

This Notice is not to be construed as a commitment by the Government to enter into a contract. This Notice is subject to revision or cancellation at any time.

There will be no exchange of funds between NASA and the selected partner(s) for responses prepared against this Notice of Partnering Opportunity or the L-DREAM proposal development. That is, NASA will not be responsible for any costs incurred by respondents in preparing a response to this Notice or by the selected partner in preparation of the L-DREAM proposal.

Details of the Early Career Initiative Opportunity

- L-DREAM proposal submission to NASA HQ selection committee: March 14th, 2025
- L-DREAM Proposal selection notification: April 25th, 2025
- Formal contract negotiation process: ~April 25th 2025 – October 1st, 2025
- L-DREAM ECI project kickoff and external collaborator performance period start: October 1, 2025
- L-DREAM ECI project end and external collaborator performance period end: September 30th, 2027

SECTION 3—RESPONSES TO NOTICE

Response Content

In the response to this partnering opportunity, interested parties should provide information regarding:

- Technical Proficiency
  - Respondents shall describe their large-scale manufacturing capabilities that can be provided to the project, with preference for roll-to-roll or large area batch magnetron sputter deposition coating manufacturing and mixed coatings manufacturing.
  - Respondents shall describe whether the above manufacturing capabilities are capable of depositing coatings onto (a) solar cell coverglass or flexible coverglass and / or (b) thermal control surface materials.
  - Respondents shall describe their research and design (R&D) capabilities with a focus on materials science and coatings and film development.
- Past Experience
  - Respondents shall describe their relevant experience in the development and manufacturing of coatings or films for extreme environments or space applications.
  - Respondents shall describe their relevant experience in the development and manufacturing of coatings or films for solar cells or thermal control surfaces.

- Business Information
    - Respondents shall describe their relevant personnel who can provide their expertise to the L-DREAM project.
    - Respondents shall provide an anticipated cost associated with providing the deliverables outlined in Section 1.

Evaluation of Response

The evaluation team will use the following factors in the selection of partner(s) for this effort. The government will evaluate the submitted response based on these factors that are listed in order of importance:

1. Technical Proficiency.
    a. The Government will evaluate the NPO Response to determine the Respondent's ability to successfully complete the deliverables outlined in Section 1.
2. Past Experience.
    a. The Government will evaluate the NPO Response to determine the relevance of the Respondent's past experience with development and manufacturing of coatings for extreme environments and space applications as well as past experience with coatings for solar cells and thermal control surfaces.
3. Business Information.
    a. The Government will evaluate the NPO Response based on quality (past experience, technical excellence) of personnel selected by the Respondent to participate in the L-DREAM project.

    b. The Government will evaluate the NPO Response based on the Respondent's ability to complete the deliverables at a reasonable and realistic cost estimate.

All responses will be treated as proprietary information and controlled as such.

Questions

All questions pertaining to this Notice must be submitted to the Contract Specialist in writing via email at hunter.c.butkovic@nasa.gov. Questions are due by 5:00 pm ET on March 10th, 2025. If the government chooses to respond to submitted questions, they will be posted at a later date as attachment to this notice.

Response Instructions

The response is limited to a two (2) page written document that fully addresses the evaluation criteria in not less than 12-point font, submitted as an electronic PDF document. Excluded from the page count are the cover letter, title pages, table of contents, resumes, and an acronym list. Responders are free to attach additional appendices that further describe their capabilities. However, only the required two (2) pages will be evaluated for selection.

The responses are due to NASA GRC no later than 5:00 pm ET on March 12th, 2025. Any responses received after this time and date will be handled as late responses in accordance with FAR 15.208. Responses to this Notice must be submitted in the form of a Portable Data Format (PDF) file. Responses shall be submitted via email to Hunter Butkovic at hunter.c.butkovic@nasa.gov.

NASA GRC anticipates completing its evaluation and selection process by March 14th, 2025. Selected partners will be contacted directly by the GRC project manager or management sponsor to develop the proposal. At that time, GRC may ask for supplemental information to your proposal.

Communication Restrictions

GRC will impose local restrictions on communications regarding this notice and the processing of responses. All communications with potential respondents will be restricted to the technical and procurement points of contact identified in the synopsis of this NPO.

Additional Information Releases

The NAICS Code and Size Standard are 541715 and 1,250, respectively.

The Government does not intend to acquire a commercial item using FAR Part 12.

NASA Clause 1852.215-84, Ombudsman, is applicable. The Center Ombudsman for this acquisition can be found at http://prod.nais.nasa.gov/pub/pub_library/Omb.html

It is the offeror's responsibility to monitor this website for the release of the solicitation and amendments (if any). Potential offerors will be responsible for downloading their own copy of the solicitation and amendments (if any).

Evaluators of responses, which are submitted under this process, may include non-US Government personnel (i.e. the Principal Investigator). Therefore, responders are hereby on notice that responses submitted hereunder may be released outside the Government but shall be released solely for the purpose of evaluation of responses as part of this selection process. All responders acknowledge acceptance of this condition by the submittal of their responses.